Exit Ticket · reflect and retrieve

Exit Ticket

Binary search may never update the bounds but still always terminate.

Use key vocabulary from memory

1. Explain bounds in your own words.

2. Explain base case in your own words.

3. Explain auxiliary space in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.

Do Now · retrieval

Paper 2 targeted reteaching

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is O(log n)?

Q2

What is a recursive base case?

Q3

What does a trace table show?

Q4

Trace search for 3 in sorted [1,3,5,7,9].

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — Locate incorrect condition or state update

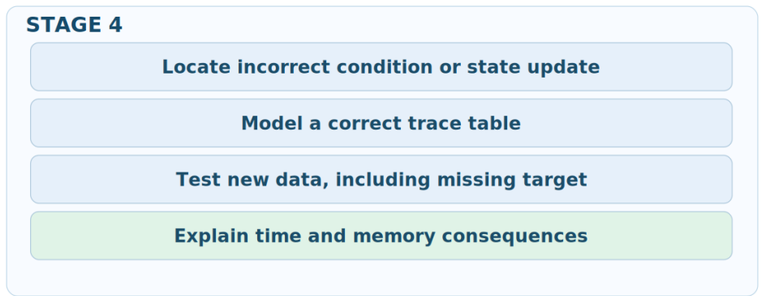
Core idea 2

Model — Model a correct trace table

Core idea 3

Apply — Test new data, including missing target

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare bubble and merge sort worst-case time/extra space; choose for million values with sufficient memory. [6]

Check that you have used the technical terms and explained your decisions.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace binary search for 4 in [2,4,6,8,10], giving bounds/midpoints. [6]

Question 2

Trace recursive SumTo(4), including base0 and return values. [6]

Check that you have used the technical terms and explained your decisions.

New Learning · applying the ideas

Core idea 4

Evaluate — Explain time and memory consequences

Vocabulary · use the correct technical terms

bounds — Limits on a search interval.

base case — Recursive stopping condition.

auxiliary space — Extra storage beyond input.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace search for 3 in [1,3,5,7,9].

Follow the worked method

1. Explain the method — The midpoint is excluded after a failed comparison, so bounds shrink.

Source model · final illustrated step

IDO · MODEL

The midpoint is excluded after a failed comparison, so bounds shrink.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the method
SumTo(2)=2+SumTo(1)=2+1+SumTo(0)=3.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.